

PREPARE Price Forecasting Report

Scope

This report summarizes the forecasting experiments run on the 'final_data_hourly' dataset for the 15-day forecasting horizon across onion, potato, tomato, and wheat.

1. Baseline Models Tried

- Strict per-crop 'HistGradientBoostingRegressor' baseline.
- Naive baselines:
 - current price
 - 7-day rolling mean
 - 28-day rolling mean
- Tabular target-shift and normalization variants:
 - 'delta_current'
 - 'delta_current + series_mean_center'
 - 'series_zscore'
- Clustered tabular model:
 - mandi clusters from mean arrival, arrival variability, and mean price
 - cluster ID used as an input feature
- Graph models:
 - GraphWaveNet
 - GraphWaveNet+
 - SAGMM-inspired graph mixture model
 - final GAT-GRU graph model with correlation-heavy edges

2. Strict Baseline Results

- Onion: 'R2 = 0.6281'
- Potato: 'R2 = 0.9255'
- Tomato: 'R2 = 0.2113'
- Wheat: 'R2 = 0.5915'

3. Naive Baseline Results

Best naive baseline for each crop:

- Onion: 'current_price = 0.7247'
- Potato: 'current_price = 0.9007'
- Tomato: 'roll_mean_28 = 0.4063'
- Wheat: 'roll_mean_7 = 0.6043'

Interpretation:

- The strict model beat naive only for potato.
- Onion, tomato, and wheat required a better problem formulation.

4. Tabular Variant Results

'delta_current'

- Onion: '0.6256'
- Potato: '0.9256'
- Tomato: '0.2615'
- Wheat: '0.6212'

'delta_current + series_mean_center'

- Onion: '0.7257'
- Potato: '0.9229'
- Tomato: '0.2674'
- Wheat: '0.6381'

'series_zscore'

- Onion: '0.7917'
- Potato: '0.8474'
- Tomato: '-0.1414'
- Wheat: '0.0696'

Interpretation:

- 'delta_current + series_mean_center' was the strongest general tabular improvement.
- 'series_zscore' helped onion but hurt the other crops.

5. Clustered Tabular Results

- Onion: '0.8078'
- Potato: '0.9047'
- Tomato: '0.3366'
- Wheat: '0.5633'

Interpretation:

- Clustering gave a major improvement for onion.
- It improved tomato over the strict baseline.
- It was worse than the best non-clustered variants for potato and wheat.

6. Early Graph Models

Tried:

- GraphWaveNet

- GraphWaveNet+
- SAGMM-inspired model

Outcome:

- All early graph models performed poorly and were not competitive with the tabular baselines.

Main reasons identified:

- weak edge construction
- thinner feature set than the tabular pipeline
- direct graph forecasting instead of anchored residual forecasting
- evaluation mismatch before restricting to observed validation rows

7. Final Graph Model

Final graph model:

- GAT-GRU mandi graph model
- one node per mandi series
- richer tabular-style lag, rolling, state, national, weather, and calendar features
- correlation + geo + state + cluster edges
- residual prediction around a crop-specific anchor

Crop-specific anchors:

- Onion: 'delta_current'
- Potato: 'delta_current'
- Tomato: 'delta_roll28'
- Wheat: 'delta_roll7'

Reason:

- each crop used the strongest simple anchor found from the naive baseline experiments

8. Final Graph Results

- Onion: '0.8544'
- Potato: '0.9287'
- Tomato: '0.4051'
- Wheat: '0.6057'

Additional graph diagnostics:

- Onion 'best_alpha = 0.75'
- Potato 'best_alpha = 0.85'
- Tomato 'best_alpha = 0.0'
- Wheat 'best_alpha = 0.1'

Interpretation:

- Onion and potato used the graph correction strongly.
- Tomato fell back almost entirely to the 28-day rolling anchor.
- Wheat used only a very small graph correction.

9. Best Current Result by Crop

- Onion: graph GAT-GRU '0.8544'
- Potato: graph GAT-GRU '0.9287'
- Tomato: naive 'roll_mean_28 = 0.4063' and graph '0.4051' are effectively tied
- Wheat: tabular 'delta_current + series_mean_center = 0.6381'

10. Main Conclusions

- Graph modeling now clearly improves onion and slightly improves potato.
- Tomato remains dominated by a smoothed anchor baseline.
- Wheat improves over the strict baseline with graph, but the best tabular model is still stronger.
- The key change that made graph modeling viable was switching to anchored residual prediction with richer features and correlation-based edges.

11. Remaining Work

- Controlled graph ablations:
 - anchor-only
 - temporal-only
 - geo-only
 - correlation-only
 - shuffled-graph
- These are needed to isolate exactly why graph structure helps onion and potato.